MIL-R-21530/1D (WEPS)
26 February 1962
Superseding
MIL-R-21530/1C (WEPS)
29 August 1961

MILITARY SPECIFICATION
RESOLVER, ELECTRICAL, AC
FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
TYPE 11R2N4r100a

1. SCOPE

1.1 This specification establishes the requirements for a Resolver, Electrical, AC, Type 11R2N4r100a, hereinafter termed resolver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21530	Resolvers, Electrical, AC, General Specification for

DRAWINGS

Bureau of Naval Weapons

1474163	Resolver, Type 11R2N4r100a

(Copies of specifications, standards, drawings, and publications, required by the contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The resolver shall be in accordance with MIL-R-21530 with the following exceptions:

(a) The resolver specified herein shall be tested only as an RX at 400 cycles per second (cps) and as an RX and RC at 10 KC.

(b) Resolver shall not be tested for salt water corrosion.

3.1.1 Testing frequency shall be as specified by purchase requisition, contract, or order.

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3.2 Design and construction. - The resolver shall be of the design, construction and physical dimensions as specified on Dwg. 1474163.

3.3 Specific characteristic requirements. - The resolver shall have specific characteristic requirements as follows:

3.3.1 Specific characteristic requirements at 400 cps.

Input voltage	26 volts root mean square (v rms) ± 10 percent
Test voltage	26 v rms ± 2 percent
Test frequency	400 cps ± 0.5 percent
Frequency range	400 and 10,000 cps
Primary	Rotor
Temperature rise	20 degrees centigrade (C), max
Friction torque, max	15 gram centimeters at room temp. 25 gram centimeters at - 55 and + 85 degrees C
Rotor impedance, Zro	See note
Stator impedance, Zso	See note
Stator impedance, Zss	See note
Rotor impedance, Zrs	See note
Total null voltage, max	20 millivolts (mv) at room temp. 30 mv at -55 and + 85 degrees C.
Angular error versus voltage	20 seconds, max
Angular error versus frequency	20 seconds, max
Angular error versus vibration	90 seconds max
Angular error versus humidity	45 seconds max
Transformer ratio (K_e)	See note
Phase shift, rotor to stator	7.0 ± 2 degrees
Phase shift versus rotation	3.5 degrees, max
Phase shift versus voltage	0.5 degrees, max
Angular error, max	180 seconds at room temp.
Null shift/paragraph 3.5.8 of MIL-R-21530	180 seconds at - 55 and + 85 degrees C.
Angular interval	10 degrees
Impedance balance	60 Z-sec, max (Res. load only)
Primary current, open circuited	0.145 amperes, nominal
Primary power, open circuited	0.76 watts, nominal
Sensitivity	0.206 volts per degree
Weight	4.7 ounces
Rotor moment of inertia	2 gram centimeters squared

NOTE: Requirements shall be as specified on Dwg. 1474163.

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3.3.2. Specific characteristic requirements at 10 KC

Input voltage	40 v ± 10 percent
Test voltage	40 v ± 2 percent
Test frequency	10,000 cps ± 0.5 percent
Rotor impedance, Zro	See note
Stator impedance, Zso	See note
Rotor impedance, Zrs	See note
Stator impedance, Zss	See note
Total null voltage, max	78 mv at room temp; 115 mv at - 85 and + 85 degrees C
Fund null voltage (max)	58 mv at room temp.
Transformation ratio (K_e)	0.462 ± 3 percent
Phase shift, rotor to stator	0.4 ± 1 degree
Primary current, open circuited	0.026 amperes, nominal
Angular error (max)	5 minutes of arc at room temp.
Impedance balance	4 minutes of arc at 150 ohm resistive loads

NOTE: Requirements shall be as specified on Dwg. 1474163.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. - Sampling and inspection shall be in accordance with MIL-R-21530.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-21530.

6. NOTES

6.1 The notes of MIL-R-21530 are applicable herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-21530/1D(WP) RESOLVER, ELECTRICAL AC FOR USE IN POLARIS FLEET BALLISTIC MISSILE

ORGANIZATION *(Of submitter)* WEAPON SYSTEM TYPE 11R2N4r100a	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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